

Aman Nindra

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EDUCATION

University of California, San Diego

Bachelor of Science in Data Science

La Jolla, CA

Expected Spring 2028

Relevant Coursework: Data Structures, Advanced Programming, Computer Organization and Assembly Language, Algorithm Design and Analysis Course, Linear Algebra, Calculus I-III, Probability & Statistics

EXPERIENCE

Undergraduate Researcher

May 2026 – Present

University of California, Merced

- Built a real-time ego-lane detection pipeline for autonomous vehicles using **LaneATT** trained on the **CULane** dataset, achieving **0.79 F1 (0.84 precision, 0.75 recall)** on the validation set; trained on a **SLURM-managed NVIDIA H100** cluster.
- Engineered a **COCO** preprocessing pipeline distilling 80 classes down to the 4 relevant to driving (person, vehicle, traffic light, stop sign) to train a **YOLOv11** detector for obstacle and traffic-control awareness.
- Integrated lane detection and object detection into a single video-inference script with frame pre/post-processing, producing annotated driving footage for qualitative evaluation of both models.

Software Engineering Intern

May 2026 – Present

Valley Institute for Sustainability, Technology & Agriculture

- Architected a machine learning training platform on a **3-node Kubernetes cluster**, including a control-plane node that manages cluster orchestration, **PostgreSQL** user data, and **5TB** of persistent storage, along with **2 GPU worker nodes** used for training. The platform enables **40+ professors** and **50+ PhD researchers** to train computer vision and NLP models.
- Built **Terraform** workspace templates that let users configure and save custom machine learning environments, including PyTorch, TensorFlow, and Conda. Users can also select hardware resources such as CPU cores, RAM, and up to **2x NVIDIA A30 GPUs** per training job. Each configuration provisions a dedicated development environment accessible through browser-based code-server or through the user's local VS Code using the **Coder** remote extension.

Software Engineering Intern

Aug. 2025 – Jan. 2026

Arrowz (Mental Health Startup)

Remote

- Restructured a **24,000-line** offline-first **Flutter/Dart** codebase into feature-based modules, eliminating duplicated legacy screens and consolidating navigation into a single reusable drawer component shared by **6** core screens.
- Built the app's complete first-run experience: a **10-step** onboarding questionnaire with per-page input validation and **Isar** local persistence, **Firestore**-authenticated routing to gate new users, and an interactive **ShowcaseView** walkthrough with database-backed progress tracking across **5+** screens.
- Shipped a goals-tracking feature end-to-end (**Isar** data model, **Provider** state management, home-screen UI) and built **4** custom date-range selector widgets (day/week/month/year) for the emotion-analytics dashboard.

PROJECTS

Self-Hosted Distributed AI Training Platform | PyTorch, Kubernetes, FastAPI, React.js, Linux

Feb. 2026 – Present

- Building a self-hosted AI training platform that lets users upload datasets (S3-style object storage), configure hyperparameters, and train models through a web interface.
- Implementing GPU-aware job scheduling with **Kubernetes** to dynamically allocate training workloads across multiple Linux machines based on real-time VRAM and compute availability.
- Enabling concurrent model training on a single GPU when memory permits, maximizing hardware utilization through resource isolation and process-level management.
- Deployed a **Raspberry Pi** as a lightweight control node hosting the website, managing job queues, and persisting uploaded datasets to attached external storage.

AI Emotion Intelligence Engine | PyTorch, Transformers, Scikit-learn, AWS SageMaker | GitHub

Jan. 2026

- Fine-tuned **BERT-base** for **28-class** multi-label emotion detection; achieved **Macro F1: 0.64–0.70** and **Jaccard (samples): 0.59–0.67**.
- Built **per-emotion threshold calibration** (0.01–0.99 grid search per label), replacing a single global cutoff and improving minority-emotion performance.
- Benchmarked against GoEmotions baselines (Macro F1 ~0.49) and achieved **Macro F1 ~0.64**, **Precision ~0.52**, and **Recall ~0.84** in 15 epochs.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C++, Dart, SQL, HTML/CSS, TailwindCSS

AI/ML: PyTorch, Scikit-Learn, OpenCV, NumPy, Pandas, YOLO, Transformers, Natural Language Processing, Computer Vision, TensorRT, ONNX Runtime

Cloud & Tools: AWS (SageMaker, S3), Kubernetes, Terraform, Slurm, Firebase, Flutter, Flask, FastAPI, Next.js, React.js, Docker, Git, Red Hat Enterprise Linux, Ubuntu